ChurnGuard

E-Commerce Customer Churn Platform

Data Exploration & Visualization Project

ITI Alexandria · Track AI · INTAKE 46

April 2026

Team Members

Ahmed ElSayed	EDA, Data Cleaning & Modeling
Amr Abdel Aziz	Power BI Dashboard
Omar Gamal	Dash Dashboard, FastAPI, Deployment & AI Agent

https://github.com/OmarGamal488/ecommerce-churn-platform
https://huggingface.co/spaces/OmarGamal48812/churnguard

Contents

1 Introduction

Customer churn — the phenomenon of customers discontinuing their relationship with a business — is one of the most critical challenges facing e-commerce companies. Acquiring a new customer costs **5–7 times more** than retaining an existing one.

> **Project Vision:** ChurnGuard is a full-stack data science platform that **predicts** which customers are at risk, **explains** why, and **recommends** what the business should do about it.

1.1 Objectives

1. **Understand** — Explore the dataset to identify patterns, quality issues, and churn drivers.
2. **Predict** — Train and evaluate multiple ML models for accurate churn prediction.
3. **Explain** — Use SHAP to make every prediction interpretable.
4. **Act** — Build interactive dashboards so business users can take data-driven actions.
5. **Deploy** — Ship the platform to a live URL accessible to anyone.

1.2 Technology Stack

Category	Technologies
Data & EDA	Python 3.11, pandas, NumPy, matplotlib, seaborn, Plotly
Machine Learning	scikit-learn, XGBoost, LightGBM, SMOTE, SHAP
Experiment Tracking	MLflow
REST API	FastAPI, Pydantic, uvicorn
Dashboard	Dash, Plotly, dash-bootstrap-components
Power BI	Power BI Desktop, DAX
AI Agent	Google Gemini 2.5 Flash, LangChain, LangGraph
Deployment	Docker, Hugging Face Spaces
Version Control	Git, GitHub, GitHub Projects

2 Dataset Description

> **Source:** Kaggle — E-Commerce Customer Churn Analysis and Prediction
> **Size:** 5,630 customers × 20 features
> **Target:** Churn (binary) — 83.16% stayed, 16.84% churned

2.1 Feature Dictionary

Feature	Type	Missing	Description
CustomerID	Integer	0%	Unique identifier (dropped)
Churn	Binary	0%	**Target:** 0=Stayed, 1=Churned
Tenure	Numeric	4.69%	Months on platform (0–61)
PreferredLoginDevice	Categorical	0%	Mobile Phone or Computer
CityTier	Categorical	0%	1 (metro), 2 (medium), 3 (small)

Feature	Type	Missing	Description
WarehouseToHome	Numeric	4.46%	Distance in km (5–36)
PreferredPaymentMode	Categorical	0%	Debit Card, Credit Card, E wallet, UPI, COD
Gender	Categorical	0%	Male or Female
HourSpendOnApp	Numeric	4.53%	Hours on app (0–5)
NumberOfDeviceRegistered	Numeric	0%	Devices registered (1–6)
PreferedOrderCat	Categorical	0%	Laptop, Mobile Phone, Fashion, Grocery, Others
SatisfactionScore	Numeric	0%	Satisfaction (1–5)
MaritalStatus	Categorical	0%	Single, Married, Divorced
NumberOfAddress	Numeric	0%	Addresses registered (1–22)
Complain	Binary	0%	**Strongest predictor:** 0=No, 1=Yes
OrderAmountHikeFromlastYear	Numeric	4.71%	% increase (11–26)
CouponUsed	Numeric	4.55%	Coupons used (0–16)
OrderCount	Numeric	4.58%	Orders placed (1–16)
DaySinceLastOrder	Numeric	5.45%	Days since last order (0–46)
CashbackAmount	Numeric	0%	Cashback received ($0–$325)

2.2 Data Quality Issues

Three categories of issues found:

1. **Missing Values:** 1,856 cells (1.65%) across 7 columns — all in the 4.5–5.5% range.
2. **Duplicate Categories:** “Phone” = “Mobile Phone”, “CC” = “Credit Card”, “COD” = “Cash on Delivery”, “Mobile” = “Mobile Phone”.
3. **Outliers:** WarehouseToHome values 126 and 127 — data entry errors (should be 26, 27).

3 Data Cleaning & Feature Engineering

3.1 Cleaning Pipeline

Implemented in `src/data_cleaning.py` as a reusable `clean()` function:

Step	Action	Detail	Impact
1	Drop CustomerID	No predictive value	20 → 19 cols
2	Merge duplicates	Phone → Mobile Phone, CC → Credit Card, etc.	3 cols fixed
3	Fix outliers	126 → 26, 127 → 27	2 values fixed
4	Median imputation	7 columns with right-skewed distributions	1,856 → 0 missing

3.2 Feature Engineering

Six new features created in `src/feature_engineering.py`:

Feature	Formula / Logic	Rationale
tenure_bucket	0–6: new, 7–12: settling, 13–24: established, 25+: loyal	Non-linear pattern
engagement_score	HourSpendOnApp $\times$ OrderCount	Combined engagement
cashback_per_order	CashbackAmount $\div$ OrderCount	Normalized metric
is_recent_buyer	1 if DaySinceLastOrder ≤ 3	Recency flag
has_multi_device	1 if Devices ≥ 4	Platform investment
is_high_spender	1 if OrderAmountHike $> 20\%$	Spending trajectory

3.3 Encoding

Label Encoding:

- Gender: Male=0, Female=1
- MaritalStatus: Single=0, Married=1, Divorced=2
- tenure_bucket: new=0 to loyal=3

One-Hot Encoding:

- PreferredLoginDevice (2 values)
- PreferredPaymentMode (5 values)
- PreferedOrderCat (5 values)

Final dataset: 5,630 rows × 30 features, zero missing values — ready for modeling.

4 Machine Learning

4.1 Experimental Setup

- **Split:** 80/20 stratified (preserves 16.84% churn in both sets)
- **Class Imbalance:** SMOTE inside pipeline — training folds only, no data leakage
- **Validation:** Stratified 5-fold cross-validation
- **Primary Metric:** F1-score (balances precision and recall for minority class)
- **Tracking:** All runs logged to MLflow with hyperparameters, metrics, and model artifacts

4.2 Model Comparison

Model	F1	AUC-ROC	AUC-PR	Accuracy
Logistic Regression	0.7823	0.9241	0.8156	0.9024
Random Forest	0.9412	0.9954	0.9821	0.9787
XGBoost	0.9523	0.9976	0.9889	0.9831
LightGBM	**0.9574**	**0.9983**	**0.9914**	**0.9858**

Table 2: Model performance on test set. LightGBM achieves the best results.

Best Model: LightGBM — F1: `0.9574` | AUC-ROC: `0.9983` | AUC-PR: `0.9914`

The pipeline (SMOTE + LightGBM) was saved as `best_model.joblib`, registered in the MLflow Model Registry, and published to Hugging Face Hub.

4.3 SHAP Explainability

SHAP (SHapley Additive exPlanations) provides both global and individual explanations:

- **Global Importance:** Complain, Tenure, NumberOfAddress, DaySinceLastOrder, CashbackAmount are the top 5 features.
- **Beeswarm Plot:** Shows direction of impact — e.g., high Complain pushes toward churn.
- **Waterfall Plots:** Individual customer explanations — "why is THIS customer at 88% risk?"

5 Business Questions & Insights

The project investigates 10 business questions spanning customer demographics, behavior, and monetary patterns.

5.1 Q1: Which tenure segments churn most?

Finding: Customers in their first 3 months churn at **36.2%** — roughly 5× the rate of any other segment. Beyond 6 months, churn stabilizes at ∼6% and reaches effectively **0% after 2 years**.

→ **Action:** Invest heavily in **onboarding experience** during the first 90 days: early incentives, personalized nudges, and proactive support. This window has the highest retention ROI of any intervention.

5.2 Q2: Does city tier affect churn?

Finding: Churn increases with city tier: **14.5% (Tier 1)** → **19.8% (Tier 2)** → **21.4% (Tier 3)**. Tier 1 holds the majority of customers (n=3,666) and retains them best, while Tier 3 loses nearly 1 in 5.

→ **Action:** Prioritize **logistics and delivery quality in Tier 3** before expanding the customer base there. Acquiring more customers in a high-churn geography without fixing root causes accelerates losses.

5.3 Q3: Which payment methods drive churn?

Finding: **Cash on Delivery (24.9%)** and **E-wallet (22.8%)** churn highest. Card payments perform best: Debit (15.4%), Credit (14.2%). CoD signals low digital commitment; E-wallet churners are likely deal-hunters switching for better cashback.

→ **Action:** Incentivize CoD users to switch to digital payments with first-time card/UPI cashback bonuses. E-wallet users need loyalty hooks beyond cashback (exclusive deals, early access).

5.4 Q4: Which product categories are linked to churn?

Finding: Mobile Phone buyers churn at 27.4% — the highest of any category and the largest segment (∼2,050 customers). Grocery is the most loyal at just **4.9%**, driven by habitual repeat purchasing.

→ **Action:** Launch post-purchase engagement for Mobile Phone buyers: accessories cross-sells, warranty upsells, trade-in programs. Use Grocery's loyalty model as a blueprint — subscription/auto-replenishment mechanics.

5.5 Q5: How does satisfaction relate to churn?

Finding — The Satisfaction Paradox: Higher satisfaction scores are associated with *higher* churn, peaking at ∼24% for score 5. Investigation reveals this is a **measurement bias**: post-complaint surveys capture momentary relief, not genuine loyalty. When controlling for complaint status, the satisfaction effect disappears entirely. **Complain is the true driver; satisfaction rides on its coattails.**

→ **Action:** Do not rely on satisfaction surveys as churn predictors. Measure **customer effort score** instead. Recognize that resolving a complaint ≠ retaining the customer — the trust damage persists.

5.6 Q6: Do complainers churn more?

Finding: Customers who filed a complaint churn at **31.7%** vs **10.9%** for non-complainers — a **3× churn multiplier** and a 20.8 percentage point gap. This is the single strongest binary predictor in the dataset.

→ **Action:** Treat every complaint as a churn alert — trigger automatic CRM retention workflow *at filing time*, not after resolution. Track time-to-resolution as a KPI. Fix top complaint categories upstream to reduce volume.

5.7 Q7: Do cashback and order hike influence loyalty?

Finding: Retained customers receive higher average cashback ($180.64 vs $160.37 for churners — a $20 gap). Churners show lower order amount growth. Neither is a strong standalone predictor, but together they paint a picture of **gradual disengagement**.

→ **Action:** Proactively increase cashback for stagnant/declining spenders. Flag customers with flat or negative order growth for 2+ months as early churn risk.

5.8 Q8–Q9: Correlation Analysis

Top Churn Correlates:

Tenure	−0.35	Longer tenure = less churn
Complain	+0.25	Complaints drive churn
DaySinceLastOrder	+0.16	Inactivity = more churn
CashbackAmount	−0.15	More cashback = less churn
SatisfactionScore	+0.11	Spurious — driven by Complain confound

Key multicollinearity: OrderCount ↔ CouponUsed ($r = 0.75$) — heavy orderers are heavy coupon users (deal-driven segment). No single feature dominates; churn is genuinely multi-factorial.

5.9 Q10: RFM Customer Profile Analysis

Finding:

- **Recency:** Misleading — churners often make a "goodbye purchase" (spike at day 0–2) before leaving. Recency is a *lagging* indicator.
- **Frequency:** The clearest RFM separator — churners are overwhelmingly **1–2 order buyers** who never developed a repeat habit.
- **Monetary:** Distributions overlap heavily; churn is not primarily a high-vs-low spender story.

→ **Action:** The critical intervention point is **between order 1 and order 3**. Design triggered re-engagement after the first purchase to push toward a second and third order — habit formation is the real retention mechanism.

6 FastAPI REST API

A microservice providing programmatic access to the ML model:

Method	Endpoint	Description
`GET`	`/health`	Health check — confirms model is loaded
`GET`	`/model-info`	Model name, type, features, pipeline steps
`POST`	`/predict`	Customer features → churn probability + risk level
`POST`	`/explain`	SHAP values for an individual prediction
`POST`	`/what-if`	Simulate feature changes, return new probability
`POST`	`/streaming/generate`	Generate synthetic customer event
`GET`	`/alerts/high-risk`	Customers with P(churn) above threshold

Table 3: FastAPI endpoints. All use Pydantic validation and CORS middleware.

7 Dash Dashboard

Built with Dash and Plotly using a **Light & Clean** theme (Inter font, Blue & Teal palette).

7.1 Tab 1: Overview

Executive summary: 6 KPI cards (Total Customers, Churn Rate, Avg Satisfaction, Avg Tenure, Complaint Rate, Stayed Count), churn distribution donut chart, churn by city tier bar chart, satisfaction histogram, and a churn rate gauge with industry-benchmark zones.

7.2 Tab 2: Customer Analysis

Four interactive cross-filters (City Tier, Login Device, Payment Mode, Marital Status) that update 6 charts simultaneously: churn by payment mode, churn by order category, satisfaction × city tier heatmap, complaint impact, gender × marital status, and device breakdown. Answers Q1, Q2, Q3, Q5.

7.3 Tab 3: Trend Analysis

Behavioral patterns: tenure distribution overlay, churn by tenure bucket, days-since-last-order box plots, order count box plots, cashback scatter plot, coupon usage bars. Answers Q4 and Q6.

7.4 Tab 4: Customer Profiles

CRM-style cards with search, risk filtering, sorting, and pagination. Each card shows demographics, behavioral metrics, color-coded churn probability, and **personalized recommendations** from 5 rules (complaint resolution, retention incentive, onboarding, re-engagement, feedback call).

7.5 Tab 5: ML Predictions

15 feature sliders for building a customer profile. Click “Predict” → real-time LightGBM prediction with a churn gauge and SHAP bar chart showing top 15 features (red = toward churn, teal = toward stay).

7.6 Tab 6: AI Churn Analyst

> Conversational AI powered by **Google Gemini 2.5 Flash** via LangChain + LangGraph.
>
> Two tools: `query_dataframe` (pandas on real data) and `predict_churn` (calls FastAPI). Users ask questions in natural language — e.g., “Which payment method has the highest churn?” or “Predict churn for a new customer who complained.”

8 Power BI Dashboard

A complementary executive dashboard built in Power BI Desktop, providing stakeholder-ready reporting across 3 pages.

8.1 Data Preparation in Power Query

Before building visuals, data was cleaned in Power Query Editor:

- **Missing values** replaced with median (Tenure=9, WarehouseToHome=14, HourSpendOnApp=3, OrderAmountHike=15, CouponUsed=1, OrderCount=2, DaySinceLastOrder=3)
- **Category standardization:** Phone → Mobile Phone, CC → Credit Card, COD → Cash on Delivery

- **Feature engineering:** Created Churn Label (Churned/Retained), Complaint Label, Tenure Group (ranges), Warehouse Distance Group
- **Data type validation:** Numerical columns set to Whole/Decimal, categorical to Text

8.2 Page 1: Overview

Visual	Detail
KPI Cards	Total Customers (5,630), Churned (948), Churn Rate (16.84%)
Donut Chart	Count of CustomerID by Churn Label — Retained ≈83%, Churned ≈17%
Column Chart	Churn Rate by Complaint Label — strong relationship between complaints and churn
Slicers	Gender and CityTier — filter all visuals interactively

8.3 Page 2: Segment Analysis

Visual	Insight
Bar Chart — Payment Mode	Highest churn: Cash on Delivery; Lowest: Credit Card
Column Chart — Marital Status	Highest churn: Single customers
Bar Chart — Order Category	Highest churn: Mobile category
Column Chart — Login Device	Computer users churn more than phone users
Column Chart — City Tier	Tier 3 cities have the highest churn
Slicer — Complaint Label	Enables deeper segmentation filtering

8.4 Page 3: Behavioral & Numerical Analysis

Visual	Insight
Avg OrderCount by Churn	Retained customers order slightly more
Avg Tenure by Churn	Retained customers stay significantly longer
Avg CashbackAmount by Churn	Retained customers receive higher cashback
Churn Rate by Tenure Group	Highest churn in early tenure (0–6 months)
Churn Rate by Warehouse Distance	Longer distance → higher churn
Churn Rate by SatisfactionScore	Lower satisfaction → higher churn (with caveats from Q5)

9 Deployment

9.1 Docker

Multi-stage Dockerfile with `docker-compose.yml` running three services:

Service	Port	Description
FastAPI	8000	ML API with health checks
Dash	8050	Dashboard (depends on FastAPI health)
Streaming	—	Background event generator

9.2 Hugging Face Spaces

Live URL: https://huggingface.co/spaces/OmarGamal48812/churnguard

Single Docker container: FastAPI (background, port 8000) + Dash (foreground, port 7860). Startup script waits for API health before launching dashboard. Gemini API key stored as encrypted Space secret.

9.3 Hugging Face Model Hub

Model URL: https://huggingface.co/OmarGamal48812/churn-prediction-lgbm

Published artifacts: `best_model.joblib`, `feature_names.joblib`, `shap_explainer.joblib` — with a model card documenting metrics, usage examples, and SHAP integration.

10 Team Contributions

Member	Responsibilities
Ahmed ElSayed	Exploratory Data Analysis notebook, data cleaning pipeline (`data_cleaning.py`), feature engineering (`feature_engineering.py`), ML model training (4 models, stratified CV), MLflow experiment tracking, SHAP explainability
Amr Abdel Aziz	Power BI dashboard (3 pages: Executive Summary, Customer Segmentation, Risk Analysis), DAX measures, slicers, cross-filtering
Omar Gamal	Dash dashboard (6 tabs), FastAPI REST API (predict, explain, what-if, streaming, alerts), AI agent (Gemini + LangChain), Docker containerization, Hugging Face deployment (Spaces + Model Hub), GitHub Projects management

11 Conclusion

ChurnGuard demonstrates a complete, production-grade approach to customer churn prediction. From a raw dataset with quality issues, we built a pipeline that cleans, engineers features, trains models, and serves predictions through interactive dashboards and a REST API.

11.1 Key Achievements

- **Model:** LightGBM — 95.74% F1, 99.83% AUC-ROC with full SHAP explainability.
- **Insights:** 10 business questions answered — complaint resolution identified as the highest-ROI action.
- **Platform:** 6-tab Dash dashboard with real-time predictions, CRM profiles, and AI analyst.
- **Deployment:** Live at a permanent URL, no installation required.

11.2 Strategic Recommendations

Based on the full analysis, we recommend the following prioritized actions:

Priority	Impact	Recommendation	Expected Outcome
1	Very High	**Fast-track complaint resolution.** Treat every complaint as a churn alert. Trigger CRM retention workflow at filing time, not after resolution.	Reduce churn among complainers from 31.7% by up to 50%
2	High	**Redesign the onboarding experience (first 90 days).** Implement welcome sequences, first-purchase rewards, and proactive check-ins during the highest-risk window.	Reduce new customer churn from 36.2% toward the 10% baseline
3	High	**Trigger re-engagement at 10+ days of inactivity.** Automated emails at day 10, discount offers at day 15, personal outreach at day 20.	Catch disengaging customers before the churn decision is final
4	Medium	**Incentivize digital payment adoption.** Offer first-time card/UPI cashback bonuses to COD users.	Reduce COD churn (24.9%) by increasing switching cost
5	Medium	**Post-purchase engagement for Mobile Phone buyers.** Cross-sell accessories, offer warranty upsells and trade-in programs within 7 days of purchase.	Convert one-time buyers into repeat customers
6	Medium	**Improve Tier 3 logistics.** Faster delivery, better tracking, and local fulfillment centers.	Reduce Tier 3 churn (21.4%) toward Tier 1 levels (14.5%)
7	Low	**Stop blanket coupon campaigns.** Use targeted promotions only for re-engagement of at-risk customers.	Reduce wasted spend on deal-seekers who churn regardless
8	Low	**Deploy weekly scoring pipeline.** Score all customers with the LightGBM model weekly and route High-risk (>60%) to immediate intervention, Medium (30–60%) to soft nudges.	Systematic, data-driven retention at scale

Table 4: Prioritized business recommendations based on data analysis and model insights.

> **Key takeaway:** The top 3 recommendations (complaint resolution, onboarding, re-engagement triggers) are all **low-cost, high-impact** interventions that can be implemented within weeks. Together, they address the three strongest churn drivers identified by both the EDA and the SHAP feature importance analysis.

11.3 Future Work

→ Real-time integration with production customer databases

→ Automated model retraining triggered by drift detection

- A/B test integration with actual campaign results to validate ROI estimates
- Multi-turn AI agent with chart generation capabilities
- Customer lifetime value (CLV) prediction to prioritize retention by revenue impact

ChurnGuard
ITI Alexandria · Track AI · INTAKE 46 · April 2026